

Workshop 3

Reminder: Midterm conflict form due soon (see Quercus)

Welcome!

Find a group to work with & say hi!

ECO 101: Principles of Microeconomics

Workshop Overview

- From individuals to a market
- Market equilibrium
- When things change (i.e. Comparative Statics)

From individuals to a market

Nothing new. Week 1's people, aggregated

W1&2: Two things you already have

- Marginal Willingness to pay: the most a buyer will pay for one more unit
 - Benefit vs Opportunity Cost, rearranged with the price on one side
- Marginal Opp. Cost: the least a seller will accept to provide one more unit
 - W1: Willingness to Accept & W2: Seller's Opp. Cost of production

Market: let buyers and sellers find each other

Used Electric Vehicle (EV) Market

- Buyers: buy one used EV or not
 - Benefit:
 - OC:
 - Resources: cash only
 - Next best alternative: One new gas-powered car

Benefit vs. OC:

- Buyer (Marginal) WTP for one used EV

Sellers are people who already own a used EV. We take each seller's min. acceptable price as given and hold it fixed today

Market: five buyers, five sellers

- Used EVs. Each person buys or sells at most one.

Buyer	B1	B2	B3	B4	B5
Most they will pay (MWTP)	\$42k	\$60k	\$36k	\$54k	\$48k

- Market Demand:

Quantity	1	2	3	4	5
Price	\$____k	\$____k	\$____k	\$____k	\$____k

- Market Supply:

Quantity	1	2	3	4	5
Price	\$____k	\$____k	\$____k	\$____k	\$____k

Seller	S1	S2	S3	S4	S5
Min. will accept (MC)	\$51k	\$33k	\$57k	\$39k	\$45k

Individuals to Market Curves



- Individuals look at a market price and decide if they want to buy (or sell) at that price
- Market demand (and supply) curves keep track of the numbers of buyers (and sellers) at different market prices, all else held fixed

Recall: Extension Question from Week 1

- Five hundred of you worked out the lowest wage that would get you out to shovel snow. Then I asked what you would have drawn if I lined all 500 up, lowest first, and counted how many would work at each wage.
- **You would have drawn a supply curve.**
- A supply curve is a queue of people, sorted by the minimum they would need to be paid to provide a good or service.
- As the price you offer them increases, more people find that the price moves the Benefit vs. Opp. Cost comparison into the accept decision category.

Worksheet Q1 (counts for grades)

Put answer on worksheet

Recall: Extension Question from Week 1

- Q1: **You would have drawn a supply curve.**
- Q2: What happens when the storm cancels everyone's classes and shifts?
- **Changing the next best alternative changes the minimum you are willing to accept to shovel. i.e. a shift of the supply curve**
- For the same price you offer, the number of people in the accept decision category changes because their Opp. Cost changed.

Summary

- Market demand (supply) adds up quantity demanded (supplied) for all consumers (producers) at the market price
- A change in market price, all else same, moves us along the curve.
- If for the same price, quantity bought or sold changes, the entire curve shifts. Shifts happen either because each person's cutoff changed, or because the set of people changed

Market Equilibrium

Quantity traded before price

W2: Two things you already have

- Gains from trade possible when there is a gap between buyers max. Willingness to Pay and seller min. acceptable price
- Price that allows for this trade to happen lies in this range
- Today: Extending logic to case with multiple different buyers and sellers
 - Price as a coordinating device

Market: five buyers, five sellers

- Used EVs. Each person buys or sells at most one.
- Market Demand

Quantity	1	2	3	4	5
Price	\$60k	\$54k	\$48k	\$42k	\$36k

- Market price:
- Market quantity:

- Market Supply

Quantity	1	2	3	4	5
Price	\$33k	\$39k	\$45k	\$51k	\$57k

Markets

- Markets don't care about names of individual producers or consumers
- Market price clears the market
 - Who wants to sell at the price?
 - Who wants to buy at the price?
 - Do the numbers of buyers = number of seller?
- Remember the assumptions behind perfectly competitive markets

When things change

Why forecasting is hard and what to keep track of

Two types of thought experiments

1. Given changes what do you predict about market price and quantity, explain
2. Given observation about changes in market price and quantity, what could explain it?
Why?

Need:

- how do demand and supply change? Shifts of the curve vs. movements along
- Net effect of demand and supply side changes

Used EV Market: after gas shock

- Used EVs, each person buys or sells at most one
- Higher gas prices raise every buyer's MWTP by \$12k.
Assume seller's min. acceptable price unchanged.

- Market Demand:

Quantity	1	2	3	4	5
Price before	\$60k	\$54k	\$48k	\$42k	\$36k
Price after	\$__k	\$__k	\$__k	\$__k	\$__k

- Market price:
- Market quantity:

- Market Supply:

Quantity	1	2	3	4	5
Price before	\$33k	\$39k	\$45k	\$51k	\$57k
Price after	\$__k	\$__k	\$__k	\$__k	\$__k

Worksheet Q2 (counts for grades)

Put answer on worksheet

Something to think about at home

Don't forget to give me your
completed worksheets

See you next workshop!
